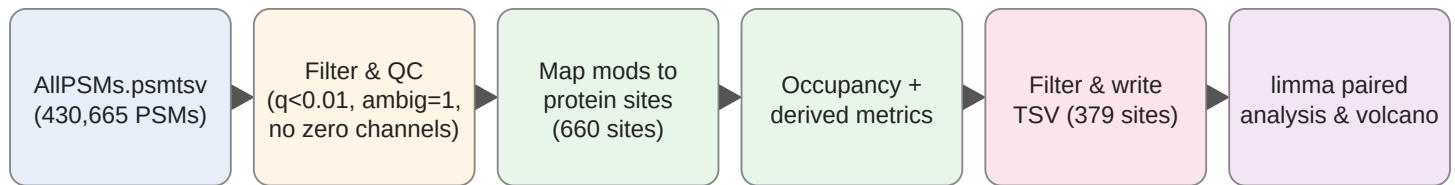


psmsToOccupancies.ipynb — Pipeline Overview

Converts MetaMorpheus PSM-level TMT intensities into per-site post-translational modification (PTM) occupancies across 18 channels (9 undrugged / 9 drugged), then runs a paired differential analysis to find sites whose occupancy shifts with drug treatment.



1. Data Cleaning

- Load **AllPSMs.psmtsv** (tab-separated, ~430k PSMs × 75 cols).
- Keep **QValue < 0.01**, **Target** hits, and **Ambiguity Level = 1** (strictest tier) → 79,137.
- Drop rows with any **zero** in the 18 TMT channels (126–135N) → **75,379**.
- Remove *insignificant* bracketed annotations (Multiplex Label, Common Variable/Fixed, **Metal**, **Artifact**) — not real PTMs.
- Strip ambiguous PSMs containing '[' — 65 residue-span rows (no sequence-level after the stricter filter).
- Unify *Phosphorylation on S* ↔ *Phosphoserine on S* as one PTM.
- Protein-group check: only **Q61029/Q61033** share peptides; coverage shows shared/unique peptides don't overlap, so groups are kept as-is.

2. Mapping to Sites

- Median-normalize channels (diagnostic only; cancels in the occupancy ratio).
- Parse **Full Sequence** with regex to recover modification name and peptide position.
- Map peptide positions → **protein positions** via the 'Start and End Residues' span (N-terminus handled specially).
- **Denominator**: sum TMT across all PSMs covering each (position, accession) — 505,524 positions.
- **Numerator**: sum TMT of modified PSMs at each site — **660 site-modifications** (each also tracks PSM count).

3. Occupancy & Output

- Per channel: occupancy = mod_intensity / total_intensity; zero-total sites are zero-filled.
- Derived metric: **mean sample intensity-weighted abundance** = occupancy × mean(total intensity) — puts low- and high-abundance sites on a common scale for stats.
- **Filter** to 0 < sum(occupancy) < 18 (drops absent/saturated sites): 660 → **379 rows**.
- Write **site_modification_occupancies.tsv**: 4 metadata cols + four 18-channel groups (mod intensity, total intensity, mean-weighted, occupancy) = **76 cols**. 25 PTM types.

4. Differential Analysis

- Input: **log₂(mean-weighted occupancy)** (closest to normal of the candidate transforms).
- **Paired design** (limma): intercept + drugged indicator + 8 mouse-pair fixed effects; eBayes variance shrinkage; **BH**-corrected p-values.
- Volcano: logFC vs $-\log_{10}(\text{adj. } p)$, marker size = mean occupancy. Top hits: H1-4 S36, Sqstm1 S24, Ctnnd1 S864, Pebp1 P11, Srp14 S45...

Assumptions & Limitations

- Channel layout alternates **undrugged**, **drugged** from channel 126; pairs are per-mouse.
- Any zero TMT channel drops the whole PSM (may bias toward high-abundance peptides).
- Ambiguous protein-group accessions (e.g. Q61029|Q61033) are **retained**; span-ambiguous rows discarded.
- All *Phospho on S* assumed phosphoserine; mean-weighting is not an ionization-efficiency correction.
- logFC is a true occupancy fold change (the per-site weight cancels in the within-pair difference); 9 pairs is small for fixed effects.